

1(a)	<table><tr><th>description</th><th>letter</th></tr><tr><td>accumulates (builds up) a high concentration of protons</td><td>B ;</td></tr><tr><td>makes triose phosphate</td><td>A ;</td></tr><tr><td>makes some chloroplast proteins</td><td>D ;</td></tr><tr><td>pumps protons</td><td>C ;</td></tr></table>	description	letter	accumulates (builds up) a high concentration of protons	B ;	makes triose phosphate	A ;	makes some chloroplast proteins	D ;	pumps protons	C ;	4
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1(b)(i)	any four from: 1 spot / extract, placed on, pencil line / base line / line of origin ; 2 repeat / concentrate, spot / extract ; 3 end / base, of paper, suspended / placed, in (named) solvent ; 4 mark the solvent front after, some time / paper removed / solvent moves ; 5 R_f value = distance moved by, spot / pigment / component / solute $\div$ distance moved by solvent (front) ; 6 compare (R_f) with known R_f values to identify, constituents / pigments ; 7 detail of method ;	4										
1(b)(ii)	any three from: 1 (pigments / they / to) absorb / capture / harvest, <u>light</u> (energy) ; 2 increase (range of) / more / different, wavelengths / colours of light, can be absorbed ; 3 increase, efficiency / rate, of, photosynthesis / light dependent reaction OR increase / broader, action spectrum ; 4 <i>ref.</i> two of: chlorophyll a / chlorophyll b / carotene / xanthophyll ;	3										